Dynamic Panels

BUSS975: Causal Inference in Financial Research

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Where we left off

- The panel lecture rested on one claim: anything about a firm that **does not change** cannot explain why its outcome changed. Demean, and the firm effect is gone.
- The price was **strict exogeneity**: today's shock must not feed into tomorrow's regressors.

- Finance is full of outcomes that depend on **their own past**: leverage, payout, cash, investment.
- Put yesterday's Y on the right-hand side and strict exogeneity is not merely doubtful. **It is false by construction.**
- This lecture: what breaks, how badly, how to fix it — and when the fix does not work.

The plan

1. **Why the past belongs on the right-hand side** — and one rule that settles every calculation.
2. **Two estimators, two wrong answers.** They err in opposite directions, which turns out to be useful.
3. **The fix: instruments from the past.** Anderson–Hsiao, then Arellano–Bond.
4. **Why the fix can fail** — and system GMM, which fails differently.
5. **In practice:** a case study from capital structure, the code, and a checklist.

- Two running questions. *Which comparison is this estimator making?* And, new this week: *which way is it wrong?*

1. Why the Past Belongs on the Right-Hand Side

Most things in finance drift

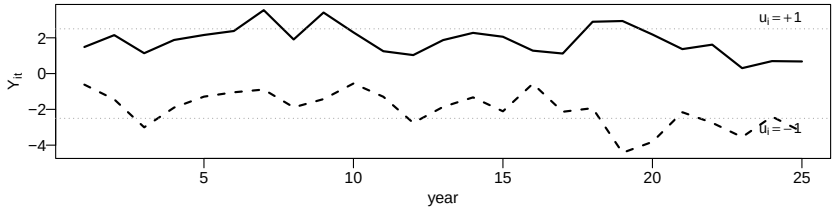
They do not jump to a target. They close part of the gap each year.

Leverage	firms close only part of the gap to target each year
Payout	managers smooth dividends, and cut only as a last resort
Cash	drawn down quickly, rebuilt slowly

$$Y_{it} \;=\; \rho\, Y_{i,t-1} + \delta D_{it} + u_i + \varepsilon_{it}, \qquad |\rho| < 1$$

- ρ **is the economics.** $1-\rho$ is the share of the gap closed per year; a shock has half-life $\ln(0.5)/\ln\rho$. At $\rho = 0.6$ that is 1.4 years; at $\rho = 0.9$, 6.6 years. **Different theories of the firm.**
- δ is now the **impact** effect. The long-run effect is $\delta/(1-\rho)$ — get ρ wrong and you get *that* wrong too.

What a firm effect does to a series



Two firms, same $\rho = 0.6$, same shocks in distribution — different u_i. Back-substitute the model into itself:

$$Y_{it} \;=\; \frac{u_i}{1-\rho} \;+\; \delta \sum_{j \geq 0} \rho^j D_{i,t-j} \;+\; \sum_{j \geq 0} \rho^j \varepsilon_{i,t-j}$$

- The firm effect shifts the **level** — not the volatility, not the persistence. Each firm wiggles around its own long-run level $u_i/(1-\rho)$.
- **Remember this picture.** An estimator that cannot tell *high because it was high last year* from *high because this firm is simply high* will get ρ wrong.

One rule settles every calculation

Y_{is} is assembled from u_i and from shocks **dated s and earlier** — nothing later. If ε_{it} is serially uncorrelated and unrelated to u_i:

$$\boxed{\mathbb{E}[Y_{is}\,\varepsilon_{it}] = 0 \;\text{ whenever }\; t > s, \qquad \mathbb{E}[Y_{is}\,\varepsilon_{is}] = \sigma_\varepsilon^2}$$

- In words: **a later shock has nothing to do with an earlier Y**; the same-date shock enters Y with coefficient one.
- Every derivation in this lecture is a **date check**: write out a product, compare the two dates, and it is either zero or σ_ε^2.
- One clause does quiet work: *serially uncorrelated*. It turns out to be the identifying assumption of the whole enterprise — and it is testable.

The problem you cannot assume away

Lag the model one period and the difficulty is on the page:

$$Y_{i,t-1} \;=\; \rho\, Y_{i,t-2} + \delta D_{i,t-1} + \underbrace{u_i}_{\text{also in today's error}} + \varepsilon_{i,t-1}$$

- Every endogenous regressor so far was endogenous because of something we **failed to measure**. This one is endogenous **by construction**.
- $Y_{i,t-1}$ contains u_i, and so does the error: **pooled OLS is biased.**
- ε_{it} moves Y_{it}, which *is* next year's regressor: strict exogeneity is false, so **fixed effects is biased too.**
- The useful part: **they fail in opposite directions.**

2. Two Estimators, Two Wrong Answers

Why OLS reads heterogeneity as persistence

Strip the model to the bone. Set $\rho = 0$ — **no persistence whatsoever**, only permanent differences between firms:

$$Y_{it} \;=\; u_i + \varepsilon_{it}$$

- A high-u_i firm is high this year **and** was high last year — not because the past caused the present, but because the firm is simply like that.
- Pooled OLS sees Y following Y and calls it persistence:

$$\operatorname{plim} \hat{\rho}_{\text{OLS}} \;=\; \frac{\operatorname{Cov}(Y_{it}, Y_{i,t-1})}{\operatorname{Var}(Y_{i,t-1})} \;=\; \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} \;>\; 0 \;=\; \rho$$

- With $\sigma_u = \sigma_\varepsilon = 1$ that is 0.5. Simulated ($T = 5$, 100,000 firms): $\hat{\rho}_{\text{OLS}} = \mathbf{0.501}$ when the truth is **zero**.
- **More years do not help.** More years of the same firm being itself is more of the same evidence.

Why demeaning manufactures mean reversion

The story. A lucky year raises $Y_{i,t-1}$ **and** the firm's average. Measured against that inflated average, the firm's other years look low — so the firm *seems* to revert even when nothing reverts.

In symbols: the firm mean contains the errors, so after demeaning

$$\underbrace{Y_{i,t-1} - \bar{Y}_{i,-1}}_{\text{regressor}} \quad \text{is correlated with} \quad \underbrace{\varepsilon_{it} - \bar{\varepsilon}_i}_{\text{error}}: \qquad \operatorname{Cov}\left(Y_{i,t-1},\, -\bar{\varepsilon}_i\right) \;<\; 0, \;\; \text{of order } \sigma_\varepsilon^2/T$$

- So $\hat{\rho}_{\mathrm{FE}}$ is biased **down**: **Nickell (1981) bias.**
- The same $\rho = 0$ experiment with fixed effects: $\hat{\rho}_{\mathrm{FE}} = -\mathbf{0.251}$, almost exactly $-1/(T-1)$. OLS invented persistence; demeaning invents **mean reversion**.
- It happens **even if** u_i **does not exist** — demeaning alone does it — and it fades only at rate $1/T$.

A derivation you can do at the board: $T = 3$

With $T = 3$ there is **one differenced observation per firm**, and $\hat{\rho}_{\text{FE}}$ is exactly least squares on $\Delta Y_{i3} = \rho \, \Delta Y_{i2} + \Delta \varepsilon_{i3}$:

$$\operatorname{plim}(\hat{\rho}_{\text{FE}} - \rho) \;=\; \frac{\mathbb{E}[\Delta Y_{i2} \, \Delta \varepsilon_{i3}]}{\mathbb{E}[(\Delta Y_{i2})^2]}$$

- **Numerator, by the rule.** Expand $(Y_{i2} - Y_{i1})(\varepsilon_{i3} - \varepsilon_{i2})$ into four products. Three pair a Y with a *later* shock: zero. One pairs Y_{i2} with its *own* shock, with a minus sign: $-\sigma_\varepsilon^2$.
- **Denominator, from the picture.** u_i cancels in a difference; what is left has variance $2\sigma_\varepsilon^2/(1+\rho)$.

$$\operatorname{plim}(\hat{\rho}_{\text{FE}} - \rho) \;=\; \frac{-\sigma_\varepsilon^2}{2\sigma_\varepsilon^2/(1+\rho)} \;=\; -\,\frac{1+\rho}{2}$$

- **The plim of $\hat{\rho}_{\text{FE}}$ is negative for every $\rho < 1$**: leverage that is genuinely 90% persistent is reported as mean-reverting. (Hansen 2022, §17.37.)
- Keep that $-\sigma_\varepsilon^2$ in mind. It comes back in Section 3.

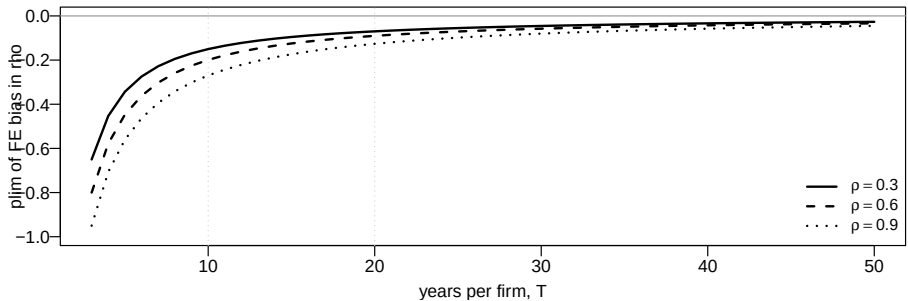
Watching both biases at once

Simulate $Y_{it} = \rho Y_{i,t-1} + u_i + \varepsilon_{it}$ with $\rho = 0.6$ and 4,000 firms, varying the number of years T.

Years per firm	$T = 3$	$T = 5$	$T = 10$	$T = 30$
Pooled OLS (biased **up**)	0.919	0.918	0.919	0.918
Fixed effects (biased **down**)	−0.189	0.147	0.396	0.543
Nickell (1981) plim, for comparison	*-0.200*	*0.153*	*0.401*	*0.542*
Truth	0.600	0.600	0.600	0.600

- **The OLS row never moves.** Heterogeneity is not resolved by more years.
- **At $T = 3$ fixed effects is negative** for a process with persistence $+0.6$ — as $-(1 + \rho)/2$ predicted.
- The FE row tracks Nickell's plim throughout: this is bias, not noise.

Does it fade with T? Slowly.



- The bias is $O(1/T)$: doubling the panel halves it. Finance panels are typically 10–20 years, often unbalanced.
- At $T = 30$ the bias is still -0.058 at $\rho = 0.6$ and -0.080 at $\rho = 0.9$. **The more persistent the series, the slower the bias leaves.**

Two wrongs that bound the right answer

Neither estimator is consistent. But you know the **sign of each bias**, so together they fence the truth in:

$$\hat{\rho}_{\text{FE}} \;<\; \rho \;<\; \hat{\rho}_{\text{OLS}}$$

- In the table at $T = 5$: $0.15 < 0.60 < 0.92$. Wide, and **honest**. A single number from either estimator would not be.
- **The standard first move** (Bond 2002): run OLS and fixed effects *before* anything clever, and keep both numbers. They are the goalposts.
- **Any estimator you trust must land between them.** One that lands outside tells you something is wrong before you interpret a single coefficient.
- A bracket is a statement about **bias**, not a confidence interval.

(A second bracket, for δ when you are unsure whether the lagged-Y or the fixed-effects model is right, is in the backup.)

3. The Fix: Instruments from the Past

The fix, in one idea: Anderson–Hsiao

Difference the model to kill u_i:

$$\Delta Y_{it} \;=\; \rho\,\Delta Y_{i,t-1} + \delta\,\Delta D_{it} + \Delta\varepsilon_{it}$$

- But $\Delta Y_{i,t-1}$ contains $\varepsilon_{i,t-1}$, and so does $\Delta\varepsilon_{it}$: the same $-\sigma_\varepsilon^2$ as the $T = 3$ derivation. **Differencing is not the fix; it is where the fixed-effects bias comes from.**
- **So go back one more lag.** $Y_{i,t-2}$ is **far enough back to be clean**, yet **close enough to still predict** $\Delta Y_{i,t-1}$.
- That is **Anderson–Hsiao (1982)**: IV on the differenced equation, instrumenting $\Delta Y_{i,t-1}$ with $Y_{i,t-2}$. It needs $T \geq 3$.
- The lagged outcome caused the problem, and a *longer* lag of the same variable solves it — no new data, only the history you already have.

One date apart: a broken regressor and a valid instrument

Check each piece against the rule, with $\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$:

	$\times\, \varepsilon_{it}$	$\times\, \varepsilon_{i,t-1}$	$\mathbb{E}[\cdot\, \Delta\varepsilon_{it}]$
$Y_{i,t-1}$	0 (later shock)	$\boldsymbol{\sigma_\varepsilon^2}$ **(same date)**	$-\sigma_\varepsilon^2$
$Y_{i,t-2}$	0 (later shock)	0 (later shock)	0
$\Delta Y_{i,t-1} = Y_{i,t-1} - Y_{i,t-2}$			$\boldsymbol{-\sigma_\varepsilon^2}$

- **Validity.** The regressor inherits the one same-date term; the instrument, one year older, has none. **The difference between a broken regressor and a valid instrument is one date.**
- **Relevance.** u_i cancels inside $\Delta Y_{i,t-1}$, leaving $\mathrm{Cov}(Y_{i,t-2}, \Delta Y_{i,t-1}) = -\sigma_\varepsilon^2/(1+\rho) \neq 0$. Nonzero — but Section 4 shows it can still be weak.
- **The zero row rests on the quiet clause.** If ε is serially correlated (fit AR(1) when the truth is AR(2)), $\varepsilon_{i,t-2}$ inside $Y_{i,t-2}$ predicts $\varepsilon_{i,t-1}$ and the instrument is invalid. **Lag length is the identifying assumption** — tested two slides on.

Arellano–Bond: use every lag you have

Anderson–Hsiao uses one moment condition out of many. At date t, **every** Y dated $t-2$ or earlier passes the date check (Holtz-Eakin, Newey and Rosen 1988; Arellano and Bond 1991):

Differenced equation at	valid instruments for $\Delta Y_{i,t-1}$	how many
$t=3$	Y_{i1}	1
$t=4$	$Y_{i1},\ Y_{i2}$	2
$t=5$	$Y_{i1},\ Y_{i2},\ Y_{i3}$	3
$\vdots$		
$t=T$	$Y_{i1},\ \ldots,\ Y_{i,T-2}$	$T-2$

- A **block-diagonal** instrument matrix with $(T-1)(T-2)/2$ columns; overidentified once $T>3$, so the estimator is **GMM**.
- **Gain:** efficiency — one-instrument Anderson–Hsiao is very noisy. **Cost:** the count grows with T^2: 91 instruments at $T=15$, 406 at $T=30$.

Too many instruments

Roodman (2009): GMM with many instruments relative to N is quietly a bad estimator.

- **It overfits.** With enough instruments the first stage reproduces $\Delta Y_{i,t-1}$ almost exactly, contaminated part included, and GMM slides back toward **OLS on its own equation**. For difference GMM that is OLS on the differences — biased **down**, below even fixed effects. (For system GMM, Section 4, it is pooled OLS: biased **up**.)
- **It disarms the overidentification test.** Hansen's J becomes too weak to reject anything; a p-value of 0.99 is a warning sign, not a clean bill of health.
- **Two standard remedies.** *Cap the lag depth* (say $Y_{i,t-2}$ to $Y_{i,t-4}$). *Collapse*: one column per lag, so the count grows with T, not T^2.
- Keep the count **well below** N, and **report it**.

Two-step weights, and the tests you must report

- **One step or two.** Two-step GMM re-weights with first-step residuals: efficient, but its standard errors are **badly biased down** in finite samples. Use the **Windmeijer (2005)** correction (`twostep vce(robust)` in Stata). If one-step robust and two-step Windmeijer disagree, something is weak.
- **AR(1) in the differenced residuals — expected.** $\Delta\varepsilon_{it}$ and $\Delta\varepsilon_{i,t-1}$ share $\varepsilon_{i,t-1}$.
- **AR(2) — must be absent.** $\Delta\varepsilon_{it}$ and $\Delta\varepsilon_{i,t-2}$ share nothing *unless* ε *is serially correlated*. This is the test of the quiet clause; fail it and **every instrument is invalid**.
- **Hansen** J, with the instrument count beside it.
- **The bracket.** Difference GMM near or below $\hat{\rho}_{\text{FE}}$: weak or too many instruments — they pull the same way. Anything above $\hat{\rho}_{\text{OLS}}$: a violated assumption (Section 4 shows one).

4. Why the Fix Can Fail

Weak instruments: the first stage, in closed form

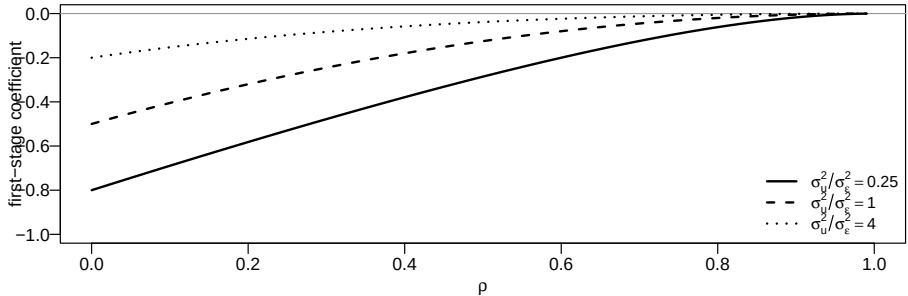
Blundell and Bond (1998) wrote the first stage, $\Delta Y_{i,t-1} = \gamma\, Y_{i,t-2} + v_{it}$, and divided covariance by variance:

$$\gamma \;=\; \frac{\mathrm{Cov}(Y_{i,t-2}, \Delta Y_{i,t-1})}{\mathrm{Var}(Y_{i,t-2})} \;=\; \frac{-\sigma_\varepsilon^2/(1+\rho)}{\sigma_u^2/(1-\rho)^2 + \sigma_\varepsilon^2/(1-\rho^2)}$$

$$\;=\; (\rho - 1)\,\frac{k}{k + \sigma_u^2/\sigma_\varepsilon^2}, \qquad k = \frac{1-\rho}{1+\rho}$$

- The covariance on top never vanishes; **the damage is all in the denominator.** The *level* $Y_{i,t-2}$ is swamped by things that say nothing about the *change*.
- **As** $\rho \to 1$, a random walk's level says nothing about its next step.
- **As** $\sigma_u^2/\sigma_\varepsilon^2$ **grows**, the level is mostly the firm's permanent height, which cancels out of the change. In the two-firm picture: the heights tell you nothing about the wiggles.
- Both are the normal case in corporate finance: **persistent series, large firm effects.**

A picture of the first stage



- At $\rho = 0.6$ with equal variances, $\gamma = -0.08$: a usable first stage. At $\rho = 0.95$, $\gamma = -0.0013$ — essentially nothing.
- Leverage, payout and cash all sit in the right-hand third of this plot. **The instrument is weakest exactly where you most need it.**

Watching it fail

Same design: 500 firms, $T = 6$, 200 replications. **Median** and **10th–90th percentile** across replications.

	$\rho = 0.6$	$\rho = 0.95$
Pooled OLS	0.92 [0.91, 0.93]	1.00 [1.00, 1.00]
Fixed effects	0.24 [0.21, 0.26]	0.47 [0.44, 0.49]
Anderson–Hsiao (one instrument)	0.59 [0.50, 0.71]	0.88 [−0.55, 2.14]
Arellano–Bond (all lags, one-step)	0.58 [0.50, 0.67]	0.60 [0.16, 0.93]

- **At $\rho = 0.6$ the fix works.** Both IV estimators sit inside the bracket, near the truth; Arellano–Bond is the tighter.
- **At $\rho = 0.95$ it does not.** Anderson–Hsiao is scattered across the number line; Arellano–Bond is pulled far down with an enormous spread — the signature of weak instruments.

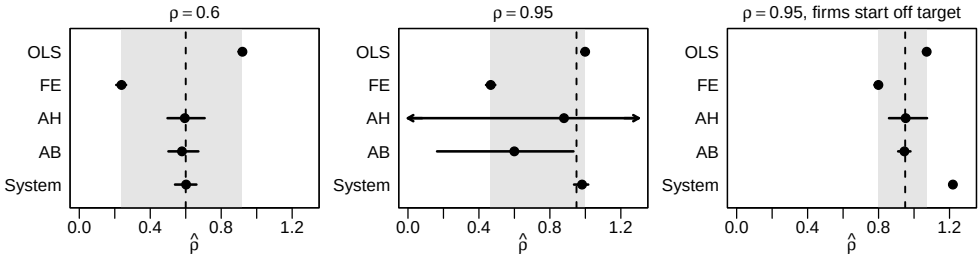
System GMM: the idea, and its price

Blundell and Bond's repair: keep the differenced equation, and **add the levels equation back**, instrumented the other way round:

$$Y_{it} = \rho\, Y_{i,t-1} + u_i + \varepsilon_{it}, \qquad \text{instrument } Y_{i,t-1} \text{ with } \Delta Y_{i,t-1}$$

- **Why it helps.** A level cannot predict a random walk's next step — but **a firm that just rose is high**. The first-stage coefficient of $Y_{i,t-1}$ on $\Delta Y_{i,t-1}$ is exactly $\frac{1}{2}$ for every ρ. It never dies.
- **What it costs.** The levels error contains u_i, so you need $\mathbb{E}[\Delta Y_{i,t-1}\, u_i] = 0$: **a firm's changes must say nothing about its permanent level.** True in the two-firm picture, *because both firms start at their own long-run levels*: **mean stationarity**.
- **When to doubt it.** Firms observed from IPO start away from their targets and converge; a panel that starts after a crisis has everyone off target together. Then rising *does* reveal a high-u_i firm, and the extra instruments are invalid.

Every estimator, on one line



Shaded: the bracket. Dashed: the truth. Dots: medians; bars: 10th–90th percentiles over 200 draws.

- $\rho = 0.6$**:** every IV estimator lands inside the bracket, near the truth.
- $\rho = 0.95$**:** the difference estimators collapse (weak instruments); system GMM holds (0.98).
- **Firms start at the average level, not their own:** Arellano–Bond recovers (0.95); system GMM lands at 1.22, **above OLS** — the bracket catches it. **Each estimator fails exactly when its own assumption does.**

Long panels: a different route

When T is large the problem is milder, and a different tool is available.

- **Bias-corrected FE.** The fixed-effects bias is $O(1/T)$ **with a known formula**: estimate FE, subtract the bias at $\hat{\rho}$, iterate (Kiviet 1995; Bruno 2005 for unbalanced panels; Stata's `xtlsdvc`). No instruments, no weak-instrument problem — but the other regressors must be **strictly exogenous**, which finance controls rarely are.
- **Long differences** — $Y_{it} - Y_{i,t-k}$ — keep the instruments strong even for persistent series (Hahn, Hausman and Kuersteiner 2007).
- **Horse race.** On finance-like panels — short, persistent, unbalanced — Flannery and Hankins (2013) find bias-corrected FE and system GMM generally reliable, and difference GMM frequently not.
- **Rule of thumb.** $T \leq 10$, ρ moderate: Arellano–Bond with the diagnostics. ρ near one: system GMM, with mean stationarity defended. $T \geq 20$ and exogenous controls: bias-corrected FE.

5. In Practice

A case study: how fast do firms adjust leverage?

The partial-adjustment model is our equation with Y = leverage and $1-\rho$ the **speed of adjustment**.

Paper	Estimator	$\hat{\rho}$	speed $1-$
Fama and French (2002)	pooled (Fama–MacBeth)	≈ 0.85	$\approx$ 7–18% a
Flannery and Rangan (2006)	fixed effects (IV for lagged leverage)	≈ 0.66	$\approx$ 34%
Lemmon, Roberts and Zender (2008)	system GMM	≈ 0.75	$\approx$ 25%
Huang and Ritter (2009)	long differencing	≈ 0.80	$\approx$ 17–23

- **The ordering is the bracket.** Fixed effects reports the *fastest* adjustment, pooled estimation the *slowest*; the IV-type estimates sit between.
- Whether firms rebalance toward targets or just drift **turned on the bias this lecture is about**: a half-life of 1.7 years and one of 5 are different theories of the firm.
- Caveat: the papers differ in samples, periods and leverage definitions. The ordering is suggestive, not a controlled horse race.

Try it: simulate, estimate, grade

```
clear all
set seed 20260202
local rho = 0.6                          // then 0.95
set obs 500
gen firm = _n
gen u = rnormal()                        // firm effect
* start each firm at its own long-run level
gen y = u/(1-`rho') + rnormal(0, 1/sqrt(1-`rho'^2))
expand 8
bysort firm: gen year = _n
xtset firm year
by firm: replace y = `rho'*y[_n-1] + u + rnormal() if _n > 1

reg   y L.y, vce(cluster firm)                    // up
xtreg y L.y, fe vce(cluster firm)                 // down
ivregress 2sls D.y (LD.y = L2.y), vce(cluster firm)
xtabond  y, lags(1) maxldep(3) twostep vce(robust)
xtdpdsys y, lags(1) maxldep(3) twostep vce(robust)
```

One draw, $\hat{\rho}$:

	0.6	0.95	0.95^{off}
OLS	0.918	0.999	1.083
FE	0.334	0.592	0.840
AH	0.596	1.077	1.004
AB	0.609	0.824	0.966
System	0.589	1.010	**1.150**

Full file: `06b_dynamic_panel_sim.do`. Set `on_target = 0` to start firms at the average level (^{off}).

One draw is noisy — read it against the figure, not on its own.

The code, on real data

```
* 1. The bracket first -- and keep both numbers
reg   y L.y x i.year, cluster(firm)
xtreg y L.y x i.year, fe cluster(firm)

* 2. Difference GMM (Arellano-Bond): two-step, Windmeijer SEs, lags 2-4 of y
xtabond y x i.year, lags(1) maxldep(3) twostep vce(robust)
estat abond                        // AR(1) expected; AR(2) must be absent

* 3. Same, with collapsed instruments and the Hansen test (Roodman's xtabond2)
ssc install xtabond2
xtabond2 y L.y x i.year, gmm(y, lag(2 4) collapse) iv(x i.year) noleveleq twostep robust
*    x predetermined rather than exogenous? replace iv(x ...) with gmm(x, lag(1 3) collapse)

* 4. System GMM -- only after you have defended mean stationarity
xtdpdsys y x i.year, lags(1) maxldep(3) twostep vce(robust)

* 5. Long T, exogenous x: bias-corrected fixed effects
ssc install xtlsdvc
xtlsdvc y x, initial(ab) bias(3) vcov(50)
```

- In R: `plm::pgmm()` with `transformation = "d"` (difference) or `"ld"` (system); `pdynmc` for the Roodman-style options.

A reporting checklist

Report	Because
$\hat{\rho}_{\text{OLS}}$ and $\hat{\rho}_{\text{FE}}$	the bracket; your estimate must sit inside it
AR(1) and AR(2) tests	AR(2) tests the identifying assumption
Hansen J, the instrument count, lag depth, collapsed or not	J is informative only when the count is small relative to N
one-step and two-step (Windmeijer) side by side	disagreement signals weakness
difference vs. system GMM, with mean stationarity defended	they rest on different assumptions
what D is assumed to be (exogenous / predetermined / endogenous)	it determines which lags of D are valid instruments

- Each row is a place where the estimator can silently fail, and a number the reader needs to decide whether it did.

Conclusion

1. **A lagged Y is endogenous by construction**, not by omission. You can read the problem off the model.
2. **One rule does every calculation**: later shocks are unrelated to earlier Y; the same-date shock enters with coefficient one. Check dates.
3. **OLS is biased up and fixed effects down.** Run both first — **the bracket is the goalposts.**
4. **The fix is a longer lag of the same variable.** $Y_{i,t-2}$ is one date too early to be contaminated. Arellano–Bond uses all such lags; cap and collapse them.
5. **The fix fails when the series is persistent or firm effects are large** — the normal case in finance. Then: system GMM if firms sit at their own long-run levels, or bias-corrected FE if T allows.

- The question to carry out: **which way is this estimator wrong?** If you can say that, you can say where the truth lies.
- **Next:** difference-in-differences — when the switching has a *date*.

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Backup

Bracketing δ: lagged Y or fixed effects?

When you are unsure **which model** is right — lagged Y, or fixed effects — and what you care about is δ, the two estimates bracket it too.

If the truth is	and you estimate	then $\hat{\delta}$ is
the lagged model	fixed effects	too **high**
the fixed effects model	the lagged model	too **low**

- As written, for $\delta > 0$; if $\delta < 0$ the directions swap. (Angrist and Pischke 2009, §§5.3–5.4.)
- Run both and report the bracket rather than picking the specification you prefer.
- **If the bracket excludes every sensible value, stop.** The model is wrong in some other way.